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A novel role of acellular hemoglobin in hemolytic thrombosis

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ABSTRACT

Background: Hemolytic thrombosis has been associated with acellular hemoglobin released from damaged red blood cells during hemolysis. However, the precise molecular mechanism underlying acellular hemoglobin-induced thrombosis remains arguable. In this study, we examined the interaction between hemoglobin and the A1 domain of von Willebrand factor (VWF), which is a critical mediator of platelet activation.

Methods: Previous studies have suggested that the interaction between hemoglobin and the A1 domain of VWF enhances VWF's hemostatic activity. We employed a multidisciplinary investigation to re-examine this interaction, and identified significant differences in binding affinity between the active and inactive forms of A1.

Results: We found that hemoglobin binds more strongly to the active A1 than the inactive form. Using hydrogen-deuterium exchange mass spectrometry, we identified the specific residues involved in this interaction, which are located on the α1-β2 and β3-α2 loops that are typically covered by the "autoinhibitory module" in the inactive A1. This observation provides a structural explanation for the differential binding affinity between the active and inactive forms of A1. We demonstrated that the binding of hemoglobin to A1 blocks the interaction between GPIbα and VWF, and inhibits VWF-mediated thrombosis in vivo. Furthermore, we found that administration of hemoglobin led to similar levels of thrombocytopenia and microthrombosis in both wildtype and VWF-deficient mice, indicating that the mechanism underlying acellular hemoglobin-induced thrombosis is VWF-independent.

Conclusions: These findings challenge the previous theory that hemoglobin-induced thrombosis occurs solely through binding with VWF, and provide evidence supporting a novel role for hemoglobin in hemolytic thrombosis.

1. Introduction

There is mounting evidence supporting a direct correlation between elevated levels of acellular hemoglobin and thrombotic complications associated with hemolysis [1–4]. However, the precise molecular mechanisms that trigger thrombosis induced by acellular hemoglobin are still not well understood. Hemoglobin is typically confined within red blood cells shuttling oxygen, but in hemolytic disorders such as sickle cell disease or during mechanical hemolysis, hemoglobin is released into the bloodstream [5]. The lifespan of acellular hemoglobin is generally short due to its rapid clearance by the scavenger protein haptoglobin, but continuous hemolysis can lead to a build-up of acellular hemoglobin in the plasma [6]. Although hemoglobin does not directly interact with platelets, high levels of acellular hemoglobin can increase the consumption of vascular nitric oxide, leading to endothelial damage and platelet activation [7–10].

Recent studies have showed that hemoglobin can also bind to von Willebrand factor (VWF), a plasma glycoprotein that initiates platelet activation in response to vascular injury [11–13]. It has been proposed that the hemoglobin/VWF complex may have an increased hemostatic activity compared to VWF alone, and the binding of acellular hemoglobin to VWF has been suggested as an important mechanism by which acellular hemoglobin stimulates thromboembolism. However, the structural basis of how hemoglobin binds to VWF and how this

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interaction enhances VWF hemostatic activity remains unclear.

A key feature in regulating the hemostatic activity of VWF is the exposure of its A1 domain. The peptide regions known as the "auto-inhibitory module (AIM)" play a significant role in modulating the exposure of A1, as demonstrated by us and others [14,15]. AIM masks the A1 domain of VWF at regions including helix α1 and loop α1-β2 and β3-α2 [14]. The unzipping of the AIM from A1 is essential for the interaction between A1 and platelet GPIbα, which is considered the initial step in VWF-mediated platelet activation. Recombinant A1 fragments carrying AIM exhibit much lower activity than those without AIM. Previous studies reported the interaction between hemoglobin and the A1 fragment carrying AIM [12]. However, the binding between hemoglobin and the A1 fragment lacking AIM has not been investigated.

In current research, we aimed to investigate 1) the binding between hemoglobin and the active form of the A1 domain of VWF and 2) molecular mechanism underlying acellular hemoglobin-induced thrombosis.

2. Methods

Most detailed methods can be found in the *Online Supplementary Methods*.

2.1. Mice

Experiments involving mice were performed in accordance with experimental protocols approved by the Institutional Animal Care and Use Committee of Soochow University. C57BL/6 J mice were used as the wildtype mice. *vwf-KO* mice were generated via the CRISPR/Cas9 technique by Biocytogen Pharmaceuticals (Beijing) Co., Ltd. Other details are listed in *Online Supplementary Methods*.

2.2. Materials

Human and mouse hemoglobin were purified from human and mouse red blood cells, respectively. Purified hemoglobin and plasma VWF were obtained by following established protocol [16,17]. Recombinant VWF A1 fragments and the ligand-binding-domain (LBD) of GPIbα were prepared as previously described [14,15,18,19]. Other materials are listed in *Online Supplementary Methods*.

2.3. Microscale thermophoresis (MST)

MST was performed using the MST-NT.115 instrument (Nanotemper, Germany). Fluorescently labeled $rVWF^{1261–1472}$ at a concentration of 31.5 nM with the Monolith NT™ Protein Labeling Kit RED-NHS according to the manufacturer's instructions was mixed with various amounts of purified human hemoglobin (4.88 nM to 160 μM) in 1× PBS buffer supplemented with 0.05 % Tween. The mixture was incubated in the dark for 30 min before being loaded into MST-grade glass capillaries. The MST measurements were performed using 40 % MST power and 50 % light-emitting diode (LED) power. The laser on and off times were 30 and 5 s, respectively. The intensities of the LED and laser were set at 40 %. The data were processed using Affinity Analysis software (v2.3) via *Kd* model fitting to estimate the *Kd* value.

2.4. Hydrogen-deuterium exchange mass spectrometry (HDX-MS) analysis

HDX-MS experiments were performed at the HDX-MS core facility of Soochow University using a high resolution Xevo-G2XS QTof equipped with a nano-UPLC, a LEAP II autosampler and a HDX manager from Waters Corp. (United States of America). The operation and data analysis are following the published protocol [14] which are detailed in the *Online Supplementary Methods.*

2.5. Platelet adhesion

Adhesion of platelets was performed largely as described [18]. In brief, a 5 mm spot on glass slide was coated with different VWF A1 fragments at 10 μg/mL in PBS at 4 °C overnight and blocked with 2 % bovine serum albumin in 1× PBS for 1 h at room temperature. The platelets were isolated from platelet-rich-plasma and re-suspended at 5 × 10^6 per mL in modified Tyrode's buffer containing 5 mM EDTA. Platelets were applied on the coated area for 30 min at 37 °C. The unattached platelets were washed away with 1× PBS buffer. The attached platelets were fixed with 4 % PFA for 10 min, permeabilized with 0.1 % Triton X-100 for 15 min and stained with 0.2 mg/mL Rhodamine-Phalloidin for 30 min. Images of platelets were acquired on an IX-81 inverted laser confocal microscope (Olympus, Japan) and analyzed by ImageJ software.

2.6. Statistical analysis

Statistical analysis was performed using Prism (GraphPad). No specific inclusion/exclusion criteria were applied from the analysis unless mentioned. Significance was determined either by two-tailed unpaired nonparametric Kolmogorov-Smirnov *t*-test or two-tailed paired parametric *t*-test with Welch's correction as indicated. Data are expressed as mean ± s.d. and differences were considered statistically significant when $p < 0.05$.

3. Results

3.1. Hemoglobin selectively binds to the active VWF-A1 domain with high binding affinity

We have showed in our previous study that the flanking residues around the VWF A1 domain termed "AIM" plays key role in regulating the activity of VWF (Fig. 1A) [14]. The recombinant VWF A1 fragment $rVWF^{1238–1493}$(carrying AIM residues) has low GPIbα-binding affinity compared to $rVWF^{1261–1472}$(carrying no AIM) [15]. To evaluate the difference in hemoglobin binding with these VWF A1 fragments, we quantified the binding affinities by ELISA. The results showed that the apparent binding affinity for hemoglobin-$rVWF^{1261–1472}$ interaction was estimated around 0.15 μM, and for hemoglobin-$rVWF^{1238–1493}$ interaction it was larger than 10 μM (Fig. 1B). The binding between hemoglobin and $rVWF^{1261–1472}$ was further confirmed by a MST measurement. The derived *Kd* value in MST was 0.083 ± 0.026 μM (Fig. 1C). Next, we performed an HDX-MS experiment to probe the binding site of hemoglobin on $rVWF^{1261–1472}$. Due to protein stability limitations, we used a mixture of $rVWF^{1261–1472}$ and hemoglobin at a concentration of 2.5 μM before a 12.5-fold dilution with a D_2O buffer for this test. We estimated that approximately 43 % of the proteins were in bound form after dilution based on a Kd value of 150 μM in this set-up. 103 peptic fragments were sequenced to cover residues between Cys1272 and Cys1458 with 5.6 redundancy (Fig. 1D). The extents of deuterium uptake for these peptides of both samples over time periods ranging from 10 to 1000 s were measured, from which the difference in relative fractional deuterium uptake for each residue was calculated and mapped to the 3D structure (Fig. 1E). As indicated, residues in α1-β2 and in β3-α2 loop are mostly involved in binding with hemoglobin. Interestingly, residues "ERLR" in loop α1-β2 showed decreased deuterium uptake at longer exchange time than shorter time compared with residues in loop β3-α2, suggesting that they may experience different binding dynamics. Notably, many of the residues involved in binding with hemoglobin are masked by AIM according to our previous study [15]. Hence, this explains why the binding affinity for hemoglobin-$rVWF^{1261–1472}$ is much stronger than that of the hemoglobin-$rVWF^{1238–1493}$ since in $rVWF^{1238–1493}$ that carries AIM these residues are masked by AIM. Above all, these results suggest that hemoglobin binds to an AIM masked region on VWF A1 domain with high affinity, and this allows

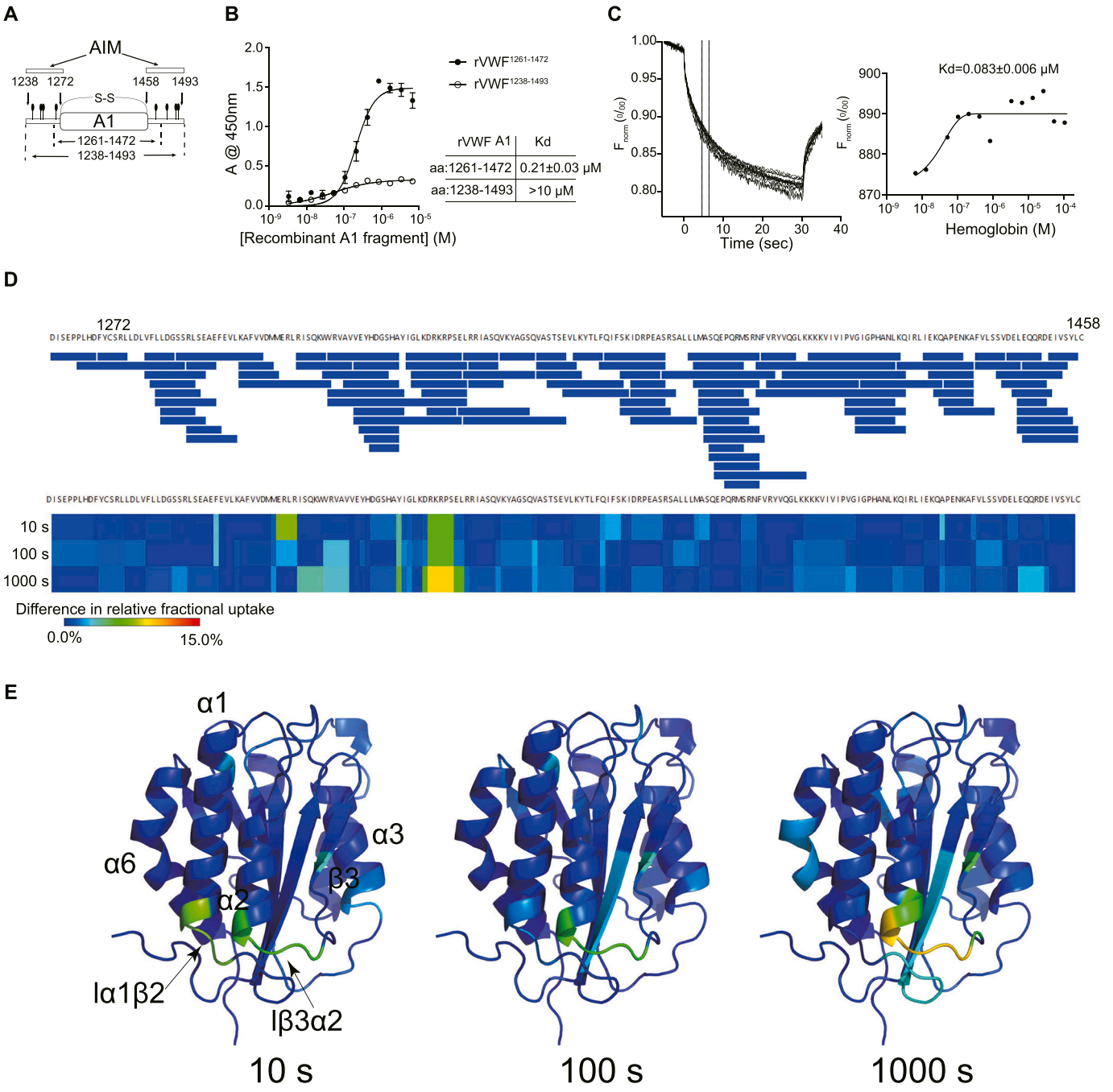


Fig. 1. Hemoglobin selectively binds to the active VWF-A1 domain with high binding affinity. (A) A schematic illustration of the AIM model and the A1 fragments from the previous study [15]. (B) Analysis of rVWF$^{1261–1472}$ (filled circle) and rVWF$^{1238–1493}$ (open circle) binding to the immobilized human hemoglobin via ELISA. The apparent dissociation constant (*Kd*) is fitted with "One Site-Total binding" equation. (C) MST curves of the binding between hemoglobin and rVWF$^{1261–1472}$. (D) The peptic fragments identified in HDX-MS characterization and the heatmap showing the averaged difference in relative fractional uptake between rVWF$^{1261–1472}$ and rVWF$^{1261–1472}$ in complex with hemoglobin. (E) The 3D projection of relative fractional deuterium uptake difference at 10, 100, and 1000 s individually to a structure of VWF A1 (PDB: 4c29). The residues are colored the same as in the heatmap.

hemoglobin to discriminate between active and inactive A1 domains.

3.2. Hemoglobin inhibits the platelet activation induced by the active VWF A1 domain

We next investigated the influence of hemoglobin on rVWF$^{1261–1472}$-induced platelet activation. rVWF$^{1261–1472}$ can spontaneously bind to platelet and induce its activation as showed by different studies [14,20]. As showed in Fig. 2A, the washed human platelets transformed from discoid shape to spread cells with pseudopodia when applied onto a rVWF$^{1261–1472}$-coated surface. In the presence of human hemoglobin at concentration of 0.38 mg/mL (6.4 μM), this morphological transformation was largely abolished (Fig. 2A and B). And platelets did not spread on the surfaces coated with rVWF$^{1238–1493}$ or hemoglobin (Fig. 2A). These results suggest that hemoglobin would inhibit the platelet morphological change induced by rVWF$^{1261–1472}$. To further test this inhibitory effect, we conducted a platelet aggregation assay. As showed in Fig. 2C, rVWF$^{1261–1472}$ induced a rapid platelet aggregation, which was inhibited by hemoglobin in a dose-dependent manner. At 0.8 μM concentration, hemoglobin postponed the platelet aggregation,

which was completely diminished by hemoglobin at 6.4 μM concentration (Fig. 2C). Consistent with this result, in a flow cytometry experiment the binding of rVWF$^{1261–1472}$ to platelets was largely inhibited by hemoglobin at concentration of 6.4 μM (Fig. 2D and S1). rVWF$^{1238–1493}$ does not bind platelets at 1 μM concentration and has no effect either on platelet aggregation or platelet activation. In consequence, the exposure of phosphatidylserine and the platelet desialylation signals were also attenuated (Figs. 2D and S1). Altogether, these results demonstrate that hemoglobin inhibits the rVWF$^{1261–1472}$-induced platelet activation.

3.3. Hemoglobin inhibits the ligand-mediated binding of full-length VWF to platelets

The observed inhibition of rVWF$^{1261–1472}$ activity by hemoglobin was unexpected and inconsistent with the previous observation that hemoglobin promotes the adhesion of platelets to the VWF-coated surface under supraphysiological conditions [12]. To test whether hemoglobin would inhibit the binding of full-length VWF to platelets under mild conditions, we investigated the influence of hemoglobin on the ligand-mediated binding of full-length VWF to platelets. Botrocetin and ristocetin are known-ligands in mediating the binding of full-length VWF with platelets as being established in numerous studies [21–23]. As showed in flow cytometry experiment, the binding between full-length VWF and platelets mediated by 2 μg/mL botrocetin was completely blocked by hemoglobin at 6.4 μM concentration (Fig. 3A). Consistently, we showed that hemoglobin could compete off the binding of the ligand-binding-domain of GPIbα (LBD) to full-length VWF in the presence of botrocetin with an IC_{50} ~ 1.85 μM (Fig. 3B). Hemoglobin at 6.4 μM concentration also inhibited the ristocetin-induced platelet aggregation, but showed no effect on the collagen-, ADP- and thrombin-induced platelet aggregations in the same condition (Fig. 3C). These results suggest that hemoglobin may specifically affect the GPIbα/VWF-A1 interaction, which is also consistent with the vicinal location between the binding site of hemoglobin and GPIbα on A1.

3.4. Transient hemoglobin administration attenuates the VWF-mediated thrombosis in mice

To verify the effect of hemoglobin on VWF-mediated platelet activation in a living environment, we launched a $FeCl_3$-induced arterial thrombosis assay that is known to be driven by GPIbα/VWF-A1 interaction [24,25]. Considering the rapid clearance of hemoglobin in plasma, a transient administration of mouse hemoglobin at 0.19 mg/mL (3.2 μM) plasma concentration was conducted only 5 min before the $FeCl_3$ stimulation. The homogeneity of the hemoglobin sample was confirmed by SDS-PAGE staining before operation (Fig. S2A). In comparison to the overwhelmingly developed thrombus in the control group

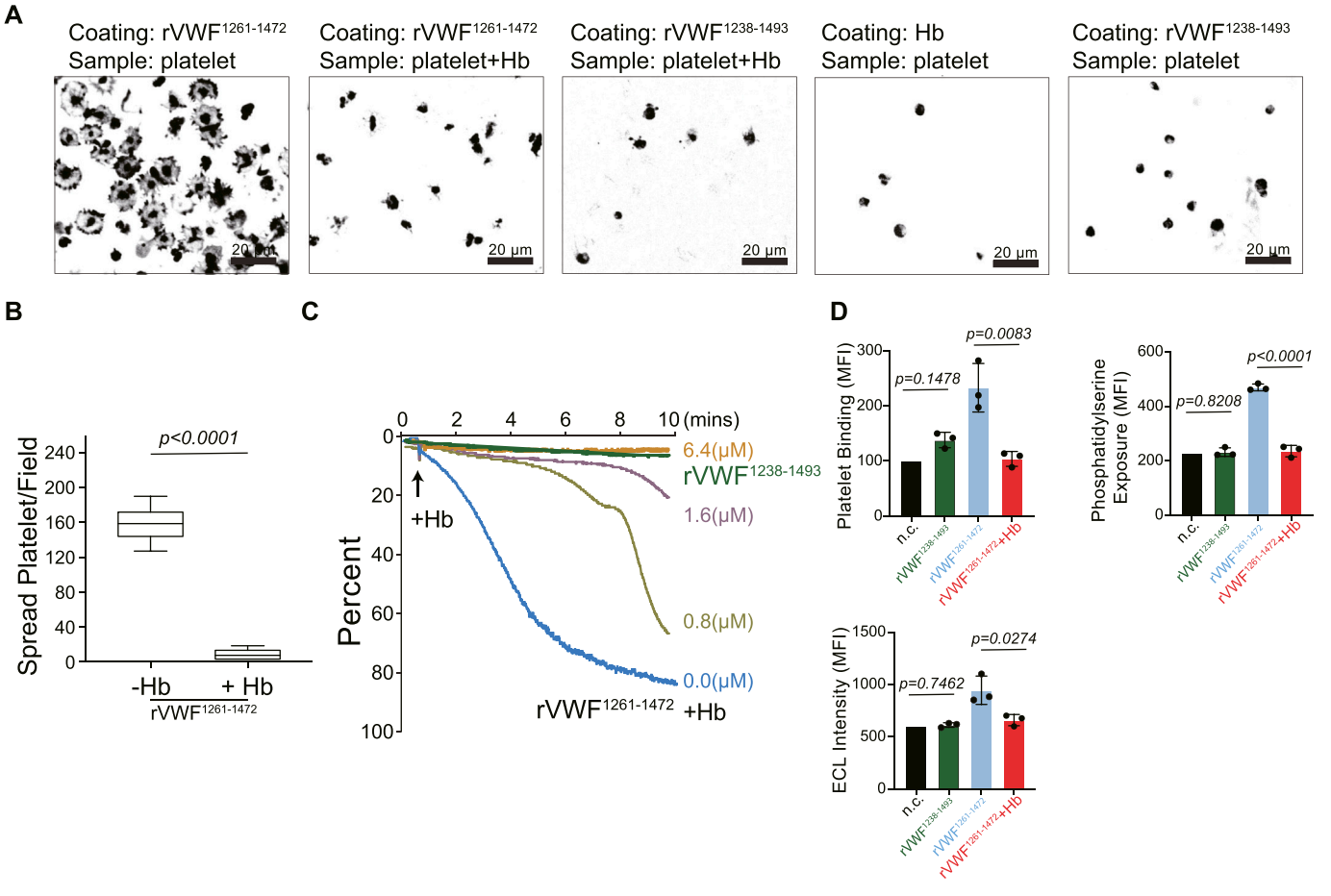


Fig. 2. Hemoglobin inhibits the platelet activation induced by the active VWF A1 domain. (A) Representative images of washed human platelets morphology on a solid surface coated with different proteins. The scale bar is 20 μM. (B) The average number of stretched platelets per observation field counted from 20 randomized images. The significance is statistically summarized using unpaired *t*-test with Welch's correction. (C) Representative platelet aggregation curves in the presence of 1 μM rVWF$^{1238–1493}$ and 0.1 μM rVWF$^{1261–1472}$ with different amount of human hemoglobin. (D) The averaged mean fluorescence intensity (n = 3) of recombinant VWF A1 domains binding to platelets and the downstream signals including phosphatidylserine exposure (indicated by PE-Annexin V) and desialylation (indicated by FITC-ECL) signals in the absence and presence of human hemoglobin at 6.4 μM concentration from flow cytometry analysis. The significance is determined using unpaired *t*-test with Welch's correction.

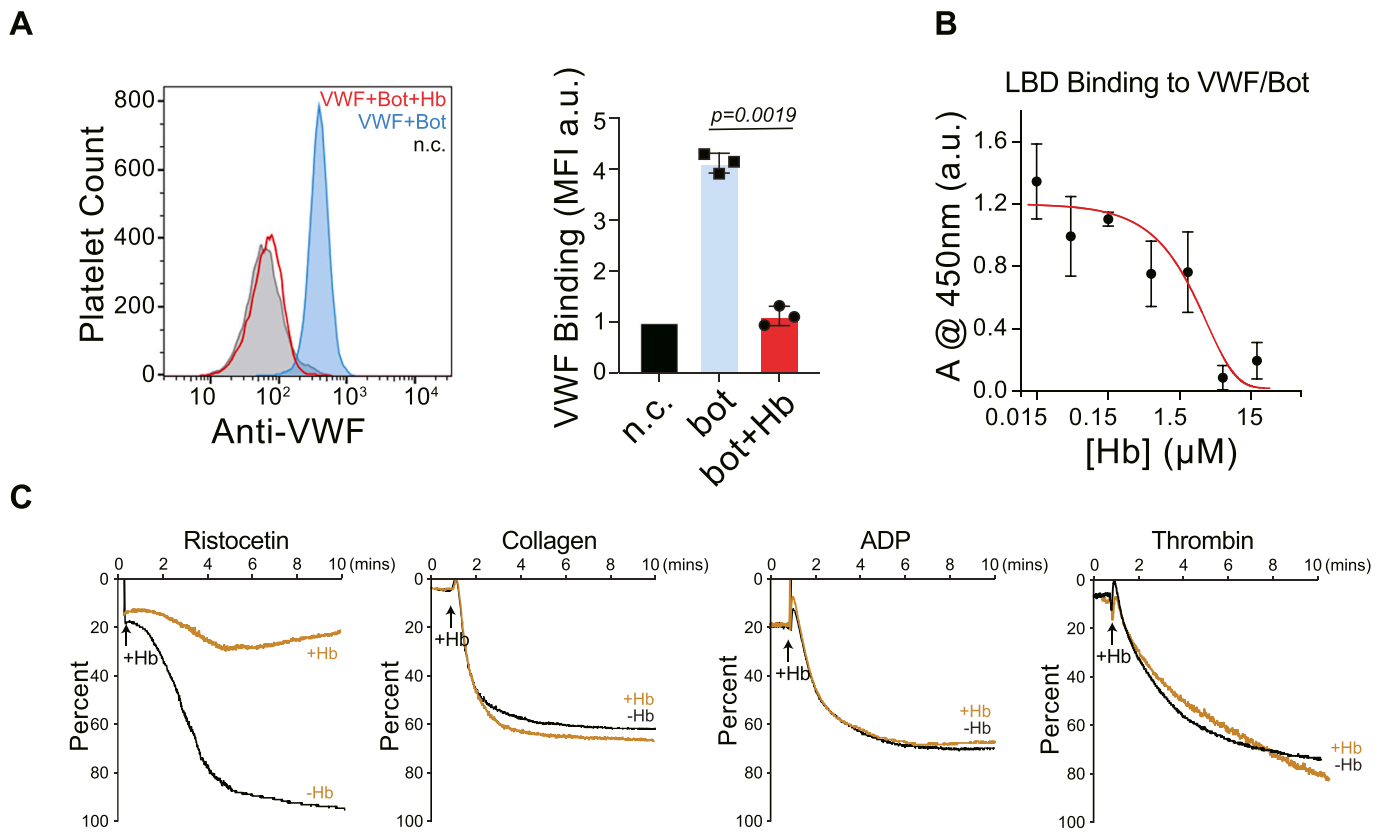


Fig. 3. Hemoglobin inhibits the ligand-mediated binding of full-length VWF to platelets. (A) The representative histogram of botrocetin-mediated binding between VWF and platelets in the absence and presence of hemoglobin at 6.4 μM concentration from flow cytometry analysis. The averaged fluorescence intensities (MFI) are compared in the bar-plot. The significance is determined using unpaired *t*-test with Welch's correction. (B) Competitive ELISA curve showing the GPIbα-LBD in binding with VWF/botrocetin in the presence of different amount of human hemoglobin. The estimated half maximal inhibitory concentration (IC_{50}) is 1.85 ± 0.45 μM. (C) Representative curves for the ristocetin-, collagen- and ADP- induced platelet aggregation in the presence and absence of hemoglobin at 6.4 μM concentration.

treated with phosphate buffer, the hemoglobin-treated group exhibited significantly reduced thrombus development (Fig. 4A). The hemin—a major heme molecule— alone group at 12 μM concentration did not show inhibitory effect (Fig. 4A). The remarkable difference among these groups in thrombus development suggests that it is the hemoglobin that blocks the VWF-mediated thrombosis (Fig. 4B). In line with this conclusion, the mice showed dramatically prolonged tail bleeding time from the hemoglobin treatment (Fig. 4C). The plasma VWF level fluctuated by <10 % during the treatment, as shown in Fig. 4D. Given that hemoglobin has a much shorter lifespan (with a $t_{1/2}$ of approximately 10 min) than VWF (with a $t_{1/2}$ of approximately 6 h), if hemoglobin were to bind to VWF, the resulting complex would likely undergo rapid clearance, leading to a decrease in VWF plasma levels. Therefore, this result suggests that hemoglobin does not interact with VWF while in circulation. Consistently, the platelet count and VWF multimer composition were not affected during the treatment (Fig. S2B and C). Since it was suggested that a hemoglobin concentration ≥ 0.5 mg/mL favors the formation of the hemoglobin-A1 complex in circulation [26], we performed a parallel study in which the hemoglobin concentration was raised to 0.95 mg/mL. The results showed that the hemoglobin at 0.95 mg/mL concentration slightly extends the thrombus development in comparison to the hemoglobin at 0.19 mg/mL concentration (Fig. S3A and B). Hemoglobin at both concentrations prolonged the tail bleeding time to a similar level (Fig. S3C). Overall, these results suggest that hemoglobin at both concentrations plays an inhibitory role in VWF-mediated thrombosis in a living environment.

3.5. *Multi-times hemoglobin administration causes thrombocytopenia and microthrombosis in VWF deficient mice*

Clinical studies have established a correlation between acellular hemoglobin and the thrombotic complication accompanying hemolysis. If the acellular hemoglobin plays an inhibitory role in VWF-mediated thrombosis, then the hemoglobin-induced thrombosis must be through a VWF-independent pathway. To test this hypothesis, we investigated the influence of acellular hemoglobin on platelet activation in a *vwf-KO* mice model. The *vwf-KO* mice have no VWF in their plasma, and exhibit a prolonged tail bleeding time but with normal platelet count (Fig. S4). Single i.v. injection of hemoglobin at 0.19 mg/mL concentration showed a half-life ~16 min as measured via western-blot (Fig. 5A). To mimic the continuous release of hemoglobin in hemolytic scenario, we conducted a multi-times i.v. injection of hemoglobin. A total of four injections of hemoglobin at 0.19 mg/mL each time were given at indicated time to both wildtype and *vwf-KO* mice in parallel (Fig. 5B). The same volume of phosphate saline buffer was given to both wildtype and *vwf-KO* mice in control group at the same time. As indicated, both wildtype and *vwf-KO* mice showed thrombocytopenia upon hemoglobin treatment, whereas the platelet count in control mice injected with phosphate saline buffer stayed normal (Fig. 5C). Inspired by this result, we further checked the formation of glomerulus microthrombosis (GMT) at 24 h after the initial injection in a parallel study. The mouse was considered GMT positive if two or more microthrombi were found in the kidney tissue sections. The results showed that three out of the four hemoglobin-treated *vwf-KO* mice were GMT positive (one out of five *vwf-KO* mice died during the hemoglobin treatment); and three out of five hemoglobin-treated

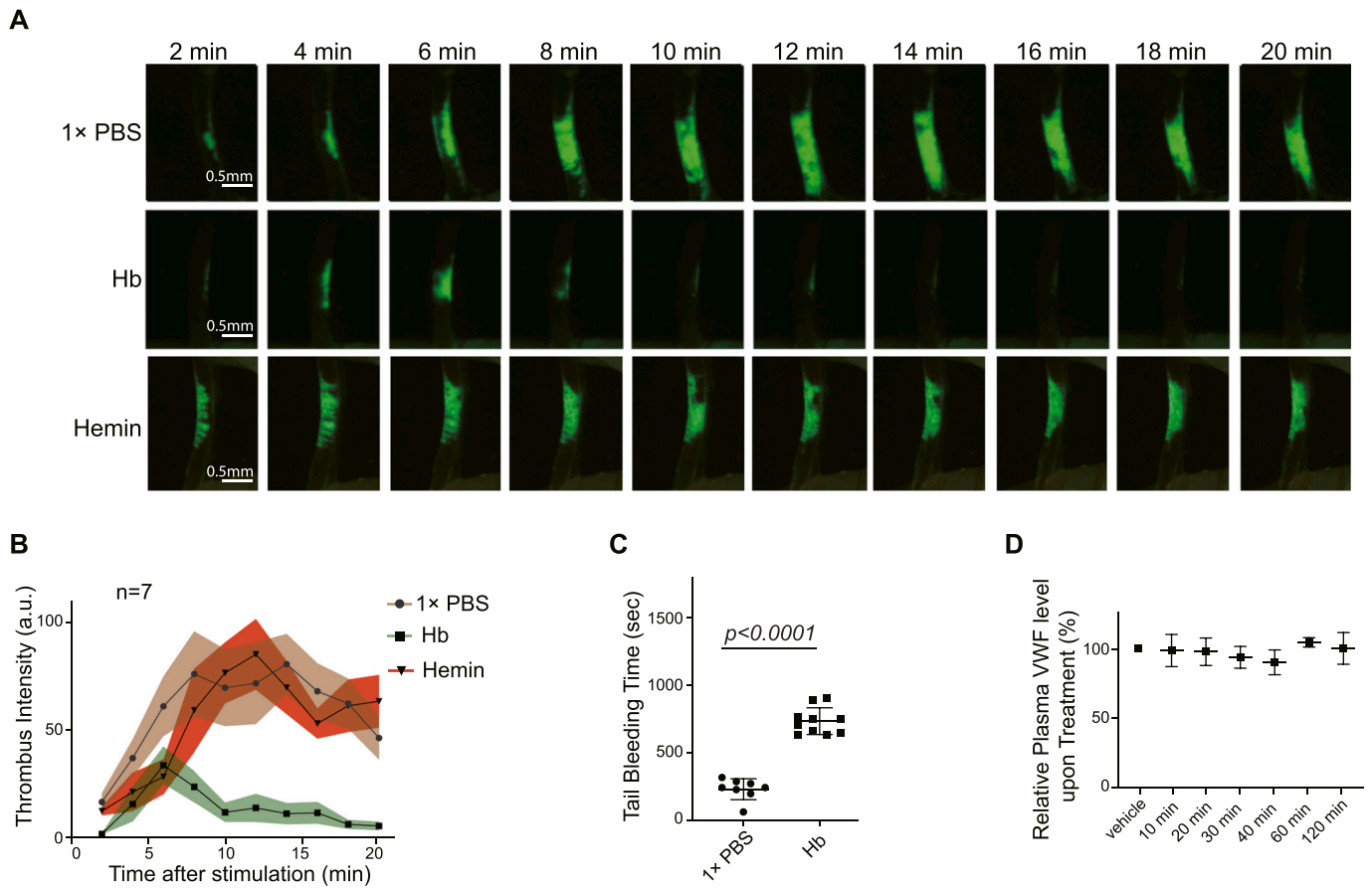


Fig. 4. Transient hemoglobin administration attenuates the VWF-mediated thrombosis in mice. (A) Development of $FeCl_3$-induced thrombosis in mice treated with phosphate-buffered saline (1× PBS), hemoglobin (Hb, 0.19 mg/mL), and hemin (12 μM). Equal number of mice ($n = 7$) are recruited for each group. Representative fluorescent stereomicroscope images of the $DioC_6$-stained thrombus every 2 min after injury. The green color reflects the accumulation of thrombi. (B) The comparison of the thrombus development among three groups. The mean value and the filled area within the standard deviation error bands of each group are shown. (C) Tail bleeding time of the mice ($n = 8$ for each group) receiving 1× PBS and hemoglobin treatments. Mean with standard deviation values are shown in scatter plot. The significance is determined using unpaired nonparametric Kolmogorov-Smirnov *t*-test. (D) Averaged plasma VWF level (determined by western-blot, n = 3) after receiving hemoglobin treatment. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

wildtype mice were GMT positive (Fig. 5D and E). In the control group treated with phosphate saline buffer, none of the mice was GMT positive (Fig. S5). These results suggest that in a VWF deficient living environment, hemoglobin may still trigger the thrombosis likely through a VWF-independent mechanism. We tested whether the inflammation-related platelet activation plays a role here. The results showed that neither the tissue factor level nor the neutrophil count were increased at 24 h after initial treatment (Fig. 5F and G). Hence, the inflammation-related platelet activation is not playing a major role in this hemoglobin-induced thrombosis.

4. Discussion

In this study, we investigated the potential interaction between hemoglobin and the VWF A1 domain, which has been proposed to play a role in increasing VWF's hemostatic activity by promoting platelet-VWF interaction and modulating hemolytic thrombosis. The research was prompted by a serendipitous finding that $rVWF^{1261–1472}$, when expressed in a hemoglobin-rich culture medium, was contaminated by a 16-kDa protein (later identified as bovine hemoglobin) and lost its activity, while $rVWF^{1238–1493}$ did not encounter the same problem. This observation led us to re-examine the role of hemoglobin in VWF-mediated thrombosis.

The ligand-binding activity of VWF A1 domain is critical in modulating the hemostatic activity of VWF. From structural perspective, the binding interfaces of botrocetin-A1 and aptamer ARC1779-A1 are similar and distal from the GPIbα-A1 binding interface [27–29]. For the latter interaction, a specific "exposure" of the binding site is required to enable the interaction [30,31]. We and others have demonstrated that the N- and C-terminal residues flanking the A1 domain cooperatively constitute an autoinhibitory module (AIM) that masks a region in A1 and impedes its binding to GPIbα [14,15,20]. As a result, recombinant A1 fragments without AIM show much stronger binding affinity to GPIbα than those carrying AIM. In the context of hemoglobin-A1 binding, Qi et al. [12] reported a ~15 μM Kd between hemoglobin and a VWF A1 fragment carrying AIM. We found a similar weak binding between hemoglobin and the inactive A1 fragment $rVWF^{1238–1493}$ that carries AIM in our study. Considering the VWF plasma concentration, the weak binding affinity indicates that even in severe hemolysis scenarios where the concentration of acellular hemoglobin is at the threshold of 0.5 mg/mL (8.4 μM), most of the plasma VWF remains unbound with hemoglobin. Indeed, our observation that the VWF level is not altered upon transient hemoglobin administration verifies the lack of interaction between hemoglobin and VWF in circulation. Otherwise, the plasma VWF level would be reduced due to the quick clearance of hemoglobin.

In this study, we have identified a novel interaction between hemoglobin and the VWF A1 domain. Our results show that the recombinant A1 fragment $rVWF^{1261–1472}$ (without AIM) has over 100 times stronger binding affinity to hemoglobin than the inactive A1 $rVWF^{1238–1493}$ carrying AIM. Interestingly, the binding site of this strong

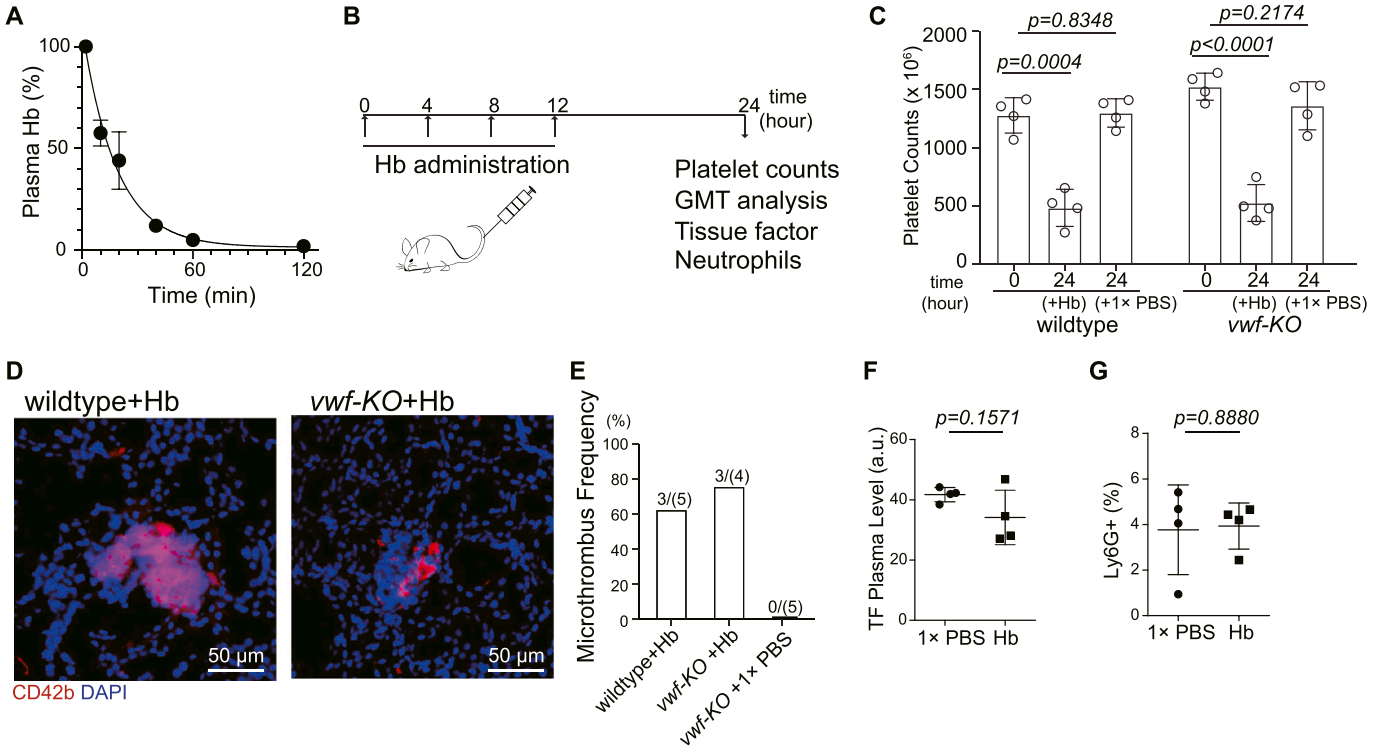


Fig. 5. Multi-times hemoglobin administration causes thrombocytopenia and microthrombosis in VWF deficient mice. (A) Metabolic kinetics of hemoglobin in *vwf-KO* mice. Purified mouse hemoglobin (0.19 mg/mL) was injected intravenously into *vwf-KO* C57BL/6 mouse, and the plasma samples were collected at different time points. The hemoglobin content in plasma was analyzed by western-blot and plotted against time. (B) Schematic illustration of the multiple-times hemoglobin administration. Purified mouse hemoglobin at 0.19 mg/mL or PBS were injected intravenously at noted time points into wildtype and *vwf-KO* C57BL/6 mouse. The platelet counts, the GMT, tissue factor and percentage of neutrophils were analyzed after 24 h from the initial injection. (C) The platelet counts comparison. The significance is determined using unpaired *t*-test with Welch's correction. (D) Representative confocal fluorescence microscope images of GMT in wildtype and *vwf-KO* C57BL/6 mouse after 24 h from the initial hemoglobin injection. The tissue samples were stained by anti-CD42b (red) and DAPI (blue). The scale bar is 50 μM. (E) The frequency of mice developed GMT in each group. The sample was counted as GMT positive if two or more microthrombi were found from the whole cross-sections. (F) The level of plasma tissue factor in each group ($n = 4$) at 24 h from the time of the initial injection of hemoglobin. The concentration of tissue factor was determined by western-blot. (G) The percentage of neutrophils in the white blood cells in each group (n = 4) at 24 h from the time of the initial injection of hemoglobin. The significance is determined using unpaired nonparametric Kolmogorov-Smirnov *t*-test. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

interaction involves residues in loop α1-β2 and loop β3-α2 on A1, located within the region masked by AIM. This finding explains why this strong interaction was not previously recognized by Qi et al. In line with the previously described "exposure" feature of A1, our results suggest that the interaction between hemoglobin and VWF A1 domain is similar to GPIbα-A1 binding, and is modulated by AIM.

The vicinal location of hemoglobin binding site and the GPIbα binding site on A1 provides a conceivable explanation for the observed inhibitory effect of hemoglobin in VWF-platelet association. This inhibitory effect was observed at physiological concentrations of hemoglobin in mild hemolytic scenarios, in contrast to the previously reported enhanced platelet adhesion in heavily-loaded hemoglobin (500 mg/dL) under supraphysiological shearing conditions [12]. The results also suggest a potential overlap between the heparin binding site and the hemoglobin binding site on A1, which may contribute to a similar mechanism between heparin and hemoglobin in preventing VWF-platelet binding. Finally, the study did not find evidence for a binding between hemoglobin and the A2 domain of VWF as previously proposed [32], or for the generation of ultra large VWF in circulation.

It is important to note that the experimental model used in this study, which involved intravenous injection of pure hemoglobin, has some limitations and may not completely mimic the pathology of hemoglobinemia and its sequelae in humans. While this model allowed for short-term observation of the effects of acellular hemoglobin on thrombosis, it does not replicate the continuous process of hemolysis seen in patients. Additionally, the model may miss important time windows and fluctuations in key agonists for platelet activation such as platelet-activating factor (PAF) [33], thrombin [34], thromboxane A2 (TxA2) [35], and adenosine diphosphate (ADP) [36], that are present in patients with hemolysis. Furthermore, the use of pure hemoglobin instead of a mixture from red blood cell lysis may not fully capture the contribution of other red blood cell components to platelet activation, such as sema7A. In addition, the VWF-deficient mice are known to have limitations like all other animal models and data obtained using those may be not applicable to humans. Therefore, caution should be taken when translating the findings of this study to human pathology.

The complex relationship between hemolysis and thrombosis suggests that there may be multiple mechanisms underlying the development of thrombosis in hemolytic disorders. While VWF has been proposed as a contributor to thrombosis in some cases, our study suggests that it is not essential for thrombosis in hemolysis and may even be inhibited by acellular hemoglobin. Other factors, such as platelet activation and consumption, may play a more important role in mediating hemolytic thrombosis. The variation in hemoglobinemia seen in different types of hemolytic disorders further underscores the complexity of the relationship between hemolysis and thrombosis. For example, in autoimmune hemolytic anemia (AIHA), which is associated with a high rate of thrombosis, hemoglobinemia is rarely observed [37]. The pathophysiological mechanism underlying hemolytic thrombosis may also vary depending on the underlying cause of hemolysis, such as infection or medication-induced hemolysis [38,39].

5. Conclusion

Overall, the relationship between hemolysis and thrombosis is complex and involves multiple factors, including platelet activation, VWF, and other inflammatory mediators. While VWF has been proposed to contribute to thrombosis in hemolysis, the results of this study suggest that hemoglobin may actually inhibit the VWF-mediated interaction with platelets. Therefore, further studies are needed to fully understand the pathophysiology of hemolytic thrombosis and to identify potential therapeutic targets for this complication.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.thromres.2023.05.024.

CRediT authorship contribution statement

Y. Wan, Y. Wei, C.Z., Y.L. and L.X. performed experiments and analyzed data; P.G., Z.Y. and J.Y. provided critical reagent; Q.Z. performed MST experiment; W.D. designed the study, analyzed data, and wrote the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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